Safouene Ziadi

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Final-year software engineering student focused on AI Systems , Deep Learning, and Interpretable Machine Learning. Experienced in building neural architectures from scratch using PyTorch and developing AI systems involving multi-agent workflows and computer vision. Seeking AI engineering or research internships/jobs.

Education

Higher Institute of Informatics and Mathematics — Monastir
Diploma in Software Engineering — 2024 - 2027

Higher Institute of Informatics and Mathematics — Monastir
Software Pre-Engineering Program — 2022-2024

Experience

Nachd-IT | Sousse | 2025-2026

- **AI Engineer** : Developing an AI-powered chatbot capable of interacting with databases through natural language queries and performing secure SQL-based operations dynamically.
- **AI Research Intern** : Conducted research on the integration of artificial intelligence within educational platforms, exploring AI-driven approaches to enhance learning experiences and student support systems.

We Are Technology Center | Monastir | 2025-2026

- **Machine Learning & Deep Learning Instructor** : Delivered 50+ hands-on training sessions on machine learning and deep learning concepts, guiding students through practical AI project development using Python and PyTorch.
- **C/C++ Instructor** : Taught programming fundamentals, problem-solving, and algorithmic thinking through structured C and C++ training sessions for 20+ Students.

Projects

QARAI Agent Guard | AI Safety , Agentic AI , Langchain | Github
• Built Qarai Agent Guard, a Python toolkit using ML and pattern-based detection to catch prompt injection, jailbreaks, and PII leaks in AI agents
• Designed severity-based policies (block, redact, warn, quarantine) with multilingual detection support
• Integrated it into LangChain and CrewAI via middleware and hooks to enforce security across agent I/O

P-SPINE Architecture (Research Project) | PyTorch, Deep Learning, NLP | Github
• Designed a novel neural architecture focused on improving the interpretability of word embeddings while preserving semantic representation quality
• Explored methods for making latent embedding dimensions more explainable and human-interpretable
• Conducted independent research and experimentation, with the project being developed as a personal research paper

Doctor Copilot | CrewAI , Multi-Agent AI, Python | Github
• Built an AI-powered multi-agent system designed to assist mental health professionals by analyzing patient digital behavior patterns
• Developed collaborative AI agents capable of processing, summarizing, and generating behavioral insights from digital interactions
• Focused on creating an extensible architecture for intelligent healthcare support and decision assistance

Durable Agent Runtime | Fault-Tolerant AI , Event-Sourcing , FastAPI | Github
• Built a crash-safe multi-step agent runtime using Python, FastAPI, and MySQL with event logs for seamless fault recovery
• Implemented row-level leases and stable idempotency keys to guarantee exactly-once side effects without redundant LLM calls
• Developed a deterministic replay engine, resumable SSE event streaming, and an interactive Streamlit UI for real-time observability

Skills

Programming Languages : Python ,C , C++ , Java , Javascript , SQL , PHP

Frameworks and Libraries : Pytorch , Keras/Tensorflow , Langchain , CrewAI , React , FastApi , Laravel

Tools : Docker , Git/Github , DBeaver , Mysql , Postman

Volunteer Experience

- **Vice President & Event Organizer (CPU-ISIMM CLUB)** : Coordinated technical events, managed team activities, and contributed to organizing student-focused workshops and initiatives.
- **Head of AI Committee (CPU-ISIMM CLUB)** : Led the AI committee by organizing and delivering multiple training sessions for 100+ students on machine learning, deep learning, and AI development topics.